

Electric Potential and Capacitors

Physics 2 Part 2 | Expanded conceptual notes and worked examples

Read each explanation, try the questions, and then follow the worked example. The final two pages provide a compact formula sheet and a variable / method key.

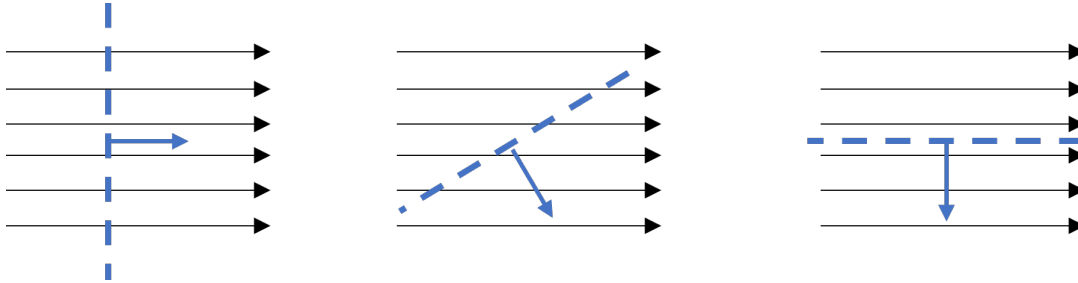
Core distinction: electric potential V is energy per charge (J/C); potential energy U is energy (J). V is scalar while E is vector. For a point charge $V=kq/r$, while a pair has $U=kq_1q_2/r$.

Constants: $k=8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0=8.85 \times 10^{-12} \text{ F/m}$; elementary charge $e=1.60 \times 10^{-19} \text{ C}$.

Gauss law review: finding E from symmetry

Electric flux measures how much electric field passes through a surface. Gauss law connects the net flux through any closed surface to the enclosed charge. It is most useful when symmetry makes E constant over a chosen Gaussian surface and fixes its direction relative to dA.

The Gaussian surface is an imaginary calculation surface; it need not match a material boundary. A charge outside the surface may create a nonzero field at its points, yet its total signed flux contribution through the closed surface cancels because field lines that enter also leave. Zero enclosed charge therefore means zero net flux, not necessarily zero field. For a conductor in electrostatic equilibrium, charges rearrange until the field inside the conducting material vanishes. Symmetry plus the right surface converts an integral into a simple product.



Course visual

Key relationships

- $\Phi_E = E A \cos(\theta)$ for a flat patch
- $\oint (\mathbf{E} \cdot d\mathbf{A}) = Q_{\text{enclosed}}/\epsilon_0$
- Outside spherical symmetry: $E = kQ/r^2$
- Long line: $E = \lambda/(2\pi\epsilon_0 r)$
- Inside electrostatic conductor material: $E = 0$

Check your understanding

- Does zero enclosed charge require zero E everywhere on the Gaussian surface?
- At a cube centered on a point charge, is E normal to every face?

Quick check: No. Zero enclosed charge gives zero net flux; the field can enter and leave. On a cube, E is not normal to every face, but Gauss law still applies.

Worked example

A long line has uniform $\lambda = 2.0 \text{ nC/m}$. Find E at $r = 0.10 \text{ m}$ using a coaxial Gaussian cylinder.

Solution. Choose radius r and length L. Symmetry gives flux $E(2\pi r L)$; enclosed charge is λL . Thus $E = \lambda/(2\pi\epsilon_0 r) = 359 \text{ N/C}$, directed radially outward.

1. Work, potential energy, and electric potential

Potential energy U belongs to an interacting system; electric potential V is energy per charge. Electrostatic force is conservative, so work depends only on the starting and ending points, not the path taken.

Electric potential gives a compact way to describe how a location affects any small test charge placed there: the test charge's energy is $U=qV$. This is why V is a property of the source charges and the location, while U also depends on the test charge. Only changes in potential are needed to calculate work; the reference value $V=0$ can be selected for convenience. Be careful with charge sign: the same potential drop changes the energy of a positive and negative particle in opposite ways.

Conservative electrostatic work depends on endpoints



Concept sketch

Key relationships

- $W_{\text{net}} = \Delta K$
- $W_{\text{electric}} = -\Delta U$
- $\Delta V = \Delta U/q = -\int \mathbf{E} \cdot d\mathbf{l}$
- Uniform field along x : $\Delta V = -E \Delta x$

Check your understanding

- Why do paths between the same endpoints give the same electrostatic work?
- A positive charge moves along $\mathbf{E}$. What happens to V ?

Quick check: Electrostatic force is conservative, so work depends on endpoints. Along $\mathbf{E}$, V decreases.

Worked example

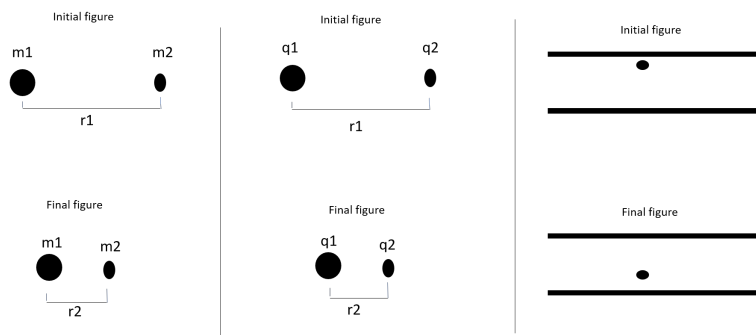
A $+2.0 \text{ microC}$ charge moves 0.15 m parallel to a uniform 400 N/C field. Find ΔV and the field work.

Solution. $\Delta V = -Ed = -(400)(0.15) = -60 \text{ V}$. $W = -q \Delta V = -(2.0 \times 10^{-6})(-60) = +1.2 \times 10^{-4} \text{ J}$.

2. Conservation of energy in electric interactions

Compare initial and final kinetic plus potential energies. A charge in electric potential has $U=qV$; a pair of point charges has $U=kq_1q_2/r$. The sign of q_1q_2 determines whether pair energy is positive or negative.

For an isolated system, a gain in kinetic energy must be balanced by an equal decrease in potential energy, and vice versa. If a particle speeds up, the potential energy of the chosen system falls. For two moving charges, both speeds may change; conservation of energy alone often supplies only one equation, so momentum conservation may also be required. Write the complete initial and final energy expressions before canceling terms. Keep signed charges in qV and in pair potential, and use absolute distance r .



Course visual

Key relationships

- $K_i + U_i = K_f + U_f$
- $\Delta K = -\Delta U = -q \Delta V$
- $U_{\text{pair}} = k q_1 q_2 / r$
- For two moving bodies, add momentum conservation if needed

Check your understanding

- What does negative pair U indicate about like or unlike charges?
- A positive charge moves to lower V . Does K rise or fall?

Quick check: Negative pair U is associated with unlike charges. A positive charge moving to lower V loses U and gains K .

Worked example

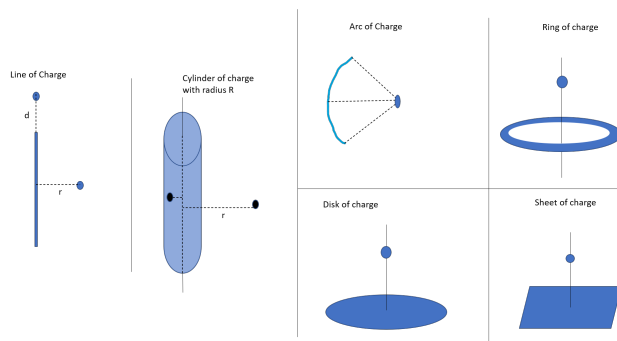
A proton at 1.00×10^5 m/s moves through a 120 V drop. Use $m_p = 1.67 \times 10^{-27}$ kg and $e = 1.60 \times 10^{-19}$ C to find final speed.

Solution. $\Delta K = -q \Delta V = 1.92 \times 10^{-17}$ J. $K_i = 8.35 \times 10^{-18}$ J, so $K_f = 2.76 \times 10^{-17}$ J and $v_f = \sqrt{2K_f/m_p} = 1.82 \times 10^5$ m/s.

3. Electric potential from charge distributions

Potential is a scalar. Use a geometry-specific expression when available and state the reference where $V=0$. Unlike E , potential contributions add without vector components.

To derive potential from a known field, integrate along a path: $\Delta V = -\int (E \cdot d\mathbf{l})$. Symmetry often reduces this to a one-dimensional integral. The integral of $1/r^2$ produces a reciprocal-radius expression, while the integral of $1/r$ produces a logarithm. For an infinite line, the potential cannot be set to zero at infinity because the logarithm does not converge there; choose a finite reference radius instead. Inside a conductor $E=0$, so the potential is constant throughout the interior and equals its surface value.



Course visual

Key relationships

- Point charge / outside spherical distribution: $V=kQ/r$
- Inside conducting sphere radius R : $V=kQ/R$
- Ring axis: $V=kQ/\sqrt{z^2+R^2}$
- Disk axis $z \geq 0$: $V=(\sigma/(2\epsilon_0))(\sqrt{z^2+R^2})-z$
- Line charge referenced to R_0 : $V(r)-V(R_0)=-2k\lambda \ln(r/R_0)$
- Far-field dipole: $V=k p \cos(\theta)/r^2$, $p=qd$

Check your understanding

- At a ring center, is V zero because E cancels?
- Why is V constant inside a conductor?

Quick check: No. Ring-element potentials add with the same sign. Inside a conductor $E=0$, so no potential difference exists between interior points.

Worked example

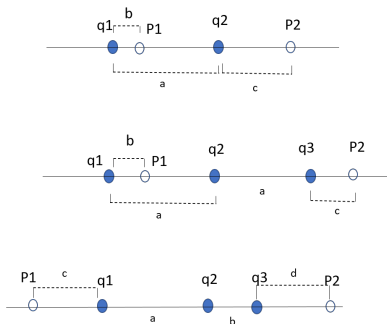
A ring of total charge $+Q$ has radius R . Find V at its center and at $z=R$ on its axis.

Solution. At the center every element is distance R , so $V=kQ/R$. At $z=R$ the distance is $R\sqrt{2}$, so $V=kQ/(R\sqrt{2})$.

4. Superposition of point-charge potentials

At P, add every charge's signed potential. Distances are positive magnitudes; the charge signs determine each term's sign. For electric field, add vectors.

A reliable setup begins with a small table of source charge, distance to P, and signed contribution kq/r . This prevents confusing a geometric distance with a signed coordinate. If the problem asks for energy of a test charge q at P, first find $V(P)$, then use $U=qV(P)$. Do not use the pair-energy formula as if it were potential: kq_1q_2/r has units of joules, while kq/r has units of volts. For several points of interest, redo distances separately for each point.



Course visual

Key relationships

- $V(P) = \sum_i k q_i / r_i$
- U of test charge q at $P = q V(P)$
- $E(P) = \text{vector } \sum_i E_i$

Check your understanding

- Can equal positive charges cancel their potentials?
- Can E be zero while V is not?

Quick check: No; positive scalar contributions add. E can cancel by vector symmetry while the scalar potential remains nonzero.

Worked example

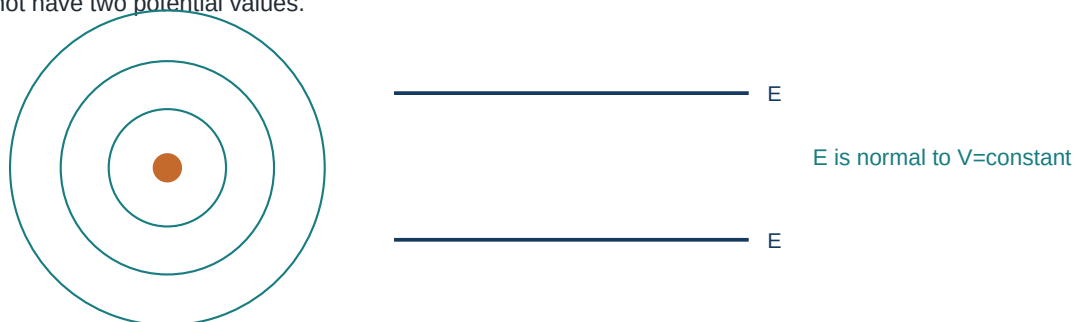
At P, $+2.0$ nC is 0.30 m away and -1.0 nC is 0.20 m away. Find $V(P)$.

Solution. $V = k[(2.0 \times 10^{-9})/0.30 - (1.0 \times 10^{-9})/0.20] = +15.0$ V.

5. Equipotential surfaces

An equipotential is a curve or surface where V is constant. Moving a charge along it requires zero electric work. E crosses equipotentials at right angles and points toward lower potential.

Equipotential maps act like contour maps of height: a charge tends to move in a direction that lowers its electric potential energy, just as an object tends downhill in gravitational potential energy. The field magnitude is large where equal-voltage contours are close together because the potential changes rapidly over a short distance. A conductor in electrostatic equilibrium is an equipotential surface. Field lines never cross one another, and distinct equipotential contours do not cross either, since a point cannot have two potential values.



Concept sketch

Key relationships

- Along an equipotential: $\Delta V=0$ and $W_{\text{electric}}=0$
- E is perpendicular to equipotential surfaces
- $|E|$ approximately $|\Delta V|/\Delta s$ for close contours

Check your understanding

- What work is done along one equipotential?
- What do tightly spaced contours suggest?

Quick check: Zero work. Tightly spaced contours indicate a large potential gradient and a strong field.

Worked example

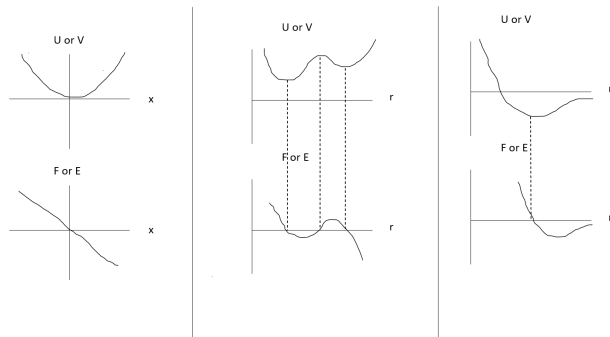
Neighboring contours differ by 20 V and are 0.050 m apart perpendicular to them. Estimate $|E|$.

Solution. $|E|$ approximately $|\Delta V|/\Delta s=20/0.050=400$ V/m; E points from higher to lower V .

6. Field as the spatial rate of change of potential

In one dimension $E_x = -dV/dx$: the field is minus the slope of $V(x)$. The same relationship connects force and potential energy: $F_x = -dU/dx$.

The minus sign encodes direction, not just magnitude. If V rises as x increases, E points toward decreasing x ; if V falls, E points toward increasing x . A steep slope means a large field, and a horizontal tangent means the component E_x is zero at that point. In multiple dimensions, the electric field is the negative gradient of potential, so it points in the direction of the fastest decrease in V . This derivative link lets you translate between graphs and motion without recalculating every force.



Course visual

Key relationships

- $E_x = -dV/dx$
- $F_x = -dU/dx$
- For a constant slope: $E_x = -\Delta V / \Delta x$

Check your understanding

- At a smooth maximum or minimum of $V(x)$, what is E_x ?
- If V slopes downward as x increases, which direction is E ?

Quick check: $E_x = 0$ at a smooth extremum. A downward slope gives E in $+x$.

Worked example

V falls linearly from 90 V at $x=0.10$ m to 30 V at $x=0.25$ m. Find E_x .

Solution. $dV/dx = (30-90)/(0.25-0.10) = -400$ V/m, so $E_x = +400$ V/m.

7. Energy of assembly and many-particle systems

Build the configuration from infinity. Each newly added charge interacts with every charge already placed. Sum over distinct pairs once; negative contributions indicate attractive pairs.

The assembly method is a bookkeeping strategy: bring in one charge at a time and calculate the work required against the existing electric field. The first charge contributes no pair-interaction energy because nothing is there to interact with. When the next charge arrives, its energy is the sum of its interactions with all earlier charges. The final answer is independent of the order of assembly, which is another consequence of conservative electrostatic forces. In atoms and molecules, this same pairwise accounting describes part of the electrostatic energy of many charged particles.



Concept sketch

Key relationships

- $U_{\text{assembly}} = \sum_{(i < j)} k q_i q_j / r_{ij}$
- Equivalent: $U = (1/2) \sum_i q_i V_{\text{other}}(r_i)$
- Each distinct pair appears once

Check your understanding

- Why does the first charge contribute no pair energy?
- Why count each pair exactly once?

Quick check: No other charge is present for the first one to interact with. Counting both (i,j) and (j,i) would duplicate every pair.

Worked example

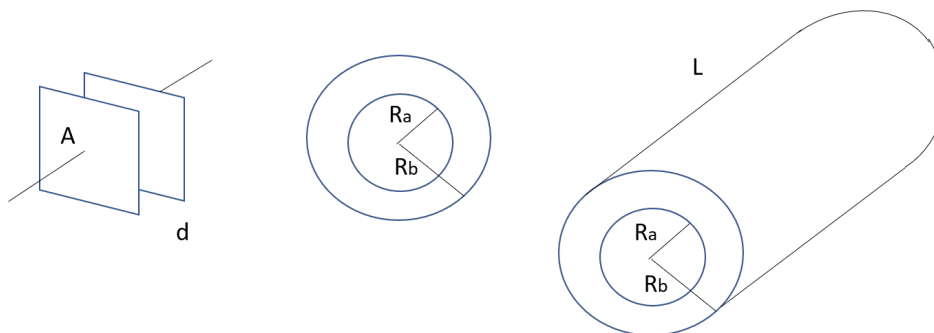
Three $+1.0 \text{ nC}$ charges are at $x=0$, 0.10 , and 0.30 m . Find U_{assembly} .

Solution. Pair distances are 0.10 , 0.20 , and 0.30 m . $U = k(1.0\text{e-}9)^2(10+5+3.33) = 1.65\text{e-}7 \text{ J}$, positive for three like charges.

8. Capacitance and capacitor geometry

Capacitance is charge magnitude per potential difference, $C=|Q|/|\Delta V|$. It depends on geometry and dielectric, not the operating Q or V for an ideal linear capacitor.

A capacitor stores equal and opposite charges on two conductors. Increasing plate area gives more room for separated charge; decreasing their separation strengthens their interaction. Both changes increase capacitance. For parallel plates the uniform-field approximation works well when the plate dimensions are much larger than the gap, though real fields fringe near edges. Spherical and cylindrical formulas apply to concentric shells and coaxial cylinders; use the correct radii and length definitions. Capacitance is measured in farads, where one farad equals one coulomb per volt.



Course visual

Key relationships

- Parallel plates: $C = \epsilon A/d$
- Concentric spherical shells: $C = 4\pi\epsilon (R_a R_b)/(R_b - R_a)$
- Coaxial cylinders: $C = 2\pi\epsilon L/\ln(R_b/R_a)$
- $\epsilon = \epsilon_r \epsilon_0$

Check your understanding

- If plate area doubles, what happens to C ?
- Why does reducing gap increase C ?

Quick check: C doubles. The smaller gap increases capacitance because the same charge creates a smaller potential difference.

Worked example

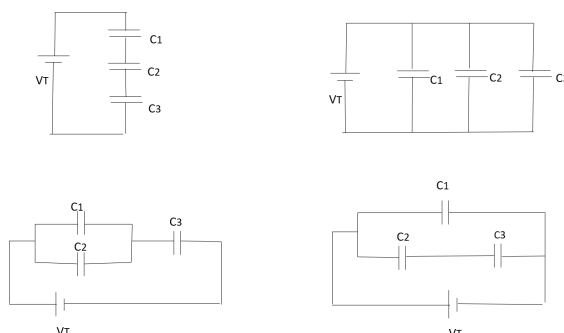
Air plates have $A=0.020 \text{ m}^2$ and $d=1.00 \text{ mm}$. Find C , then with $\epsilon_r=3.2$.

Solution. $C_0 = \epsilon_0 A/d = 1.77 \times 10^{-10} \text{ F}$. Filled capacitance is $3.2C_0 = 5.67 \times 10^{-10} \text{ F}$.

9. Capacitors in series and parallel

Parallel capacitors share voltage and their charges add. Series capacitors carry equal charge magnitude and their voltage drops add. Reduce a mixed network one group at a time, redrawing after each step.

Identify nodes before deciding whether components are in series or parallel. Two capacitors are parallel only when both terminals connect to the same two nodes; they are in series when their shared intermediate node connects to no other branch. In a series string, the smaller capacitor receives the larger voltage because $V=Q/C$ and Q is shared. The equivalent capacitance of a series group must be smaller than every individual member; a parallel equivalent must exceed every member. These comparisons are useful error checks.



Course visual

Key relationships

- Parallel: $C_{eq}=C_1+C_2+\dots$
- Series: $1/C_{eq}=1/C_1+1/C_2+\dots$
- Two in series: $C_{eq}=C_1C_2/(C_1+C_2)$

Check your understanding

- Which connection gives equal voltage?
- Why is series C_{eq} less than the smallest capacitor?

Quick check: Parallel gives equal voltage. Reciprocal capacitances add in series, making C_{eq} smaller than each member.

Worked example

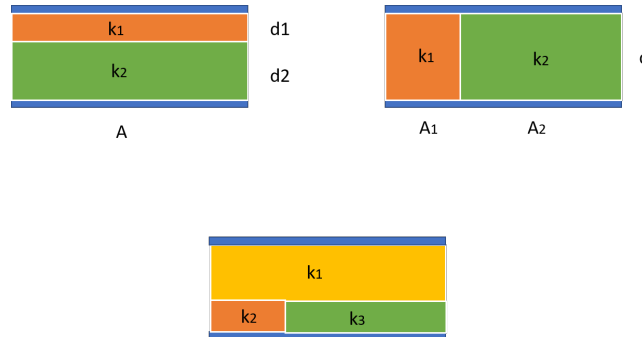
4.0 microF and 6.0 microF are parallel; this pair is in series with 5.0 microF. Find C_{eq} .

Solution. Parallel pair=10 microF. Then $C_{eq}=(10)(5)/(10+5)=3.33$ microF.

10. Dielectrics and dielectric combinations

A dielectric polarizes in the applied field. Its induced field opposes the original field and reduces the net field for fixed free charge. A full dielectric increases capacitance by ϵ_r .

Always ask whether the capacitor is isolated or still connected to a battery before deciding what changes. If isolated, no conducting path exists for free charge to enter or leave, so Q is fixed; reduced E and V accompany increased C . If battery-connected, the source maintains V , so additional charge flows onto the plates as C rises. Layer geometry matters: dielectric slabs stacked through the gap behave as series sections, while materials occupying side-by-side portions of plate area behave in parallel. Simple arithmetic or harmonic averages apply only to equal fractions.



Course visual

Key relationships

- $\epsilon_r = C_{\text{new}}/C_{\text{old}}$
- Fixed Q : $E_{\text{new}} = E_{\text{old}}/\epsilon_r$ and $V_{\text{new}} = V_{\text{old}}/\epsilon_r$
- Equal-thickness series layers: $1/\epsilon_{\text{eq}} = (1/2)(1/\epsilon_1 + 1/\epsilon_2)$
- Equal-area parallel regions: $\epsilon_{\text{eq}} = (\epsilon_1 + \epsilon_2)/2$

Check your understanding

- What stays fixed for isolated versus battery-connected capacitors?
- Why does induced E oppose the applied field?

Quick check: Isolated: Q fixed. Battery-connected: V fixed. Polarized dipoles align so their field opposes the applied field.

Worked example

An isolated capacitor has $C_0 = 47 \text{ microF}$; E changes from 250 to 230 N/C when filled. Estimate ϵ_r and C_{new} .

Solution. For fixed Q , $\epsilon_r = E_{\text{old}}/E_{\text{new}} = 250/230 = 1.087$. $C_{\text{new}} = 1.087(47) = 51.1 \text{ microF}$.

11. Energy stored in a capacitor

Charging a capacitor requires work, stored in its electric field. Choose the energy formula that matches the known variables. At fixed C , doubling Q or V quadruples U .

The factor of one-half appears because the capacitor voltage rises from zero to its final value as charge is added. The average voltage during charging is therefore half the final voltage, giving $U = (1/2)QV$. For a fixed capacitance, field energy also follows $U = (1/2)CV^2$. Energy density tells how much energy is stored per unit volume of electric field, which is useful when the field occupies a large region. A dielectric changes both the field and the energy; whether energy rises or falls depends on whether Q or V is held fixed.



field energy density: $u = \frac{1}{2} \epsilon_0 E^2$

Concept sketch

Key relationships

- $U_C = (1/2)QV = (1/2)CV^2 = Q^2/(2C)$
- Electric-field density: $u_E = (1/2)\epsilon_0 E^2$

Check your understanding

- At fixed C , what happens to U if V doubles?
- Why does U contain a one-half factor?

Quick check: U quadruples. The voltage rises during charging, so average voltage is half its final value.

Worked example

An 8.0 microF capacitor is charged to 12 V. Find Q and U .

Solution. $Q = CV = 96 \text{ microC}$. $U = (1/2)CV^2 = 5.76 \times 10^{-4} \text{ J}$.

Formula Sheet

Physics 2 Part 2 | Electric potential, fields, energy, and capacitance

Electric potential and field $V = kq/r$ $V_{\text{total}} = \sum (k q_i/r_i)$ $U_{\text{pair}} = k q_1 q_2/r$ $U = qV$ $\Delta V = -\int (E \cdot dl)$ $E_x = -dV/dx$ $F_x = -dU/dx$	Gauss law $\Phi_E = \int (E \cdot dA)$ $\Phi_E = EA \cos(\theta)$ $\Phi_E, \text{closed} = Q_{\text{enclosed}}/\epsilon_0$ $E_{\text{line}} = \lambda/(2 \pi \epsilon_0 r)$ $E_{\text{sphere, outside}} = kQ/r^2$	Capacitance $C = Q/\Delta V$ Parallel plates: $C = \epsilon A/d$ Spherical shells: $C = 4 \pi \epsilon R_a R_b / (R_b - R_a)$ Coaxial cylinders: $C = 2 \pi \epsilon L / \ln(R_b/R_a)$ $\epsilon = \epsilon_r \epsilon_0$
Energy conservation $K_i + U_i = K_f + U_f$ $\Delta K = -\Delta U = -q \Delta V$ $U_{\text{assembly}} = \sum_{(i < j)} k q_i q_j / r_{ij}$	Potential from symmetric sources Ring axis: $V = kQ/\sqrt{z^2 + R^2}$ Disk axis: $V = (\sigma/(2\epsilon_0))(\sqrt{z^2 + R^2} - z)$ Long line: $\Delta V = -2k \lambda \ln(r_2/r_1)$ Conductor sphere interior: $V = kQ/R$	Networks and stored energy Parallel capacitors: $C_{\text{eq}} = \sum (C_i)$ Series capacitors: $1/C_{\text{eq}} = \sum (1/C_i)$ $U_c = (1/2)QV = (1/2)CV^2 = Q^2/(2C)$ $u_E = (1/2)\epsilon E^2$ Dielectric: $C_{\text{new}} = \epsilon_r C_{\text{old}}$

Fast checks: volts = joules/coulomb; E = volts/meter = newtons/coulomb; capacitance in farads. Keep charge signs in V and U. Use magnitudes for distances.

Variables and When to Use Each Formula

Symbol	Meaning / unit	Use it when...
q, Q	Charge (C); q may be signed, Q often a magnitude	Use signed q in $V=kq/r$, $U=qV$, and pair-energy formulas. For capacitor plate charge, Q usually means positive magnitude.
r, r_{ij}	Distance (m) from source to point or between charges	Use a positive geometric distance. In point-source sums, find a separate r_i from every source to the same target point.
$V, \Delta V$	Electric potential ($V = J/C$); potential difference (V)	Use V when finding energy per unit charge; use ΔV between two locations for work or a field integral.
$U, \Delta U$	Potential energy (J); change in potential energy	Use $U=qV$ for a charge in a known potential; use kq_1q_2/r for a point-charge pair; sum distinct pairs for assembly energy.
E, E_x	Electric field ($N/C = V/m$); x component	Use vector addition for point-charge fields. Use $E=-dV/dx$ in one dimension or Gauss law when symmetry makes flux simple.
Φ_E, A, θ	Electric flux ($N \cdot m^2/C$), area (m^2), angle between E and area normal	Use $EA \cos(\theta)$ for a uniform field over a flat patch. For closed surfaces, integrate and use enclosed charge.
C, C_{eq}	Capacitance / equivalent capacitance ($F = C/V$)	Use $C=Q/\Delta V$ for any capacitor. Reduce networks with parallel/series rules before solving for Q or V .
ϵ, ϵ_r	Permittivity (F/m); relative dielectric constant (unitless)	Use $\epsilon=\epsilon_r \epsilon_0$ for a filled dielectric. Check whether Q is fixed (isolated) or V is fixed (battery connected).
A, d, L, R	Plate area (m^2), gap (m), cylinder length (m), geometry radii (m)	Match the geometry formula exactly: flat plates, concentric spheres, or coaxial cylinders.
λ, σ, ρ	Linear (C/m), surface (C/m^2), volume (C/m^3) charge density	Use the density matching the charge distribution. In Gauss law, enclosed charge is density times enclosed length, area, or volume.
k, ϵ_0	Coulomb constant ($N \cdot m^2/C^2$); vacuum permittivity (F/m)	Use $k=1/(4\pi\epsilon_0)$. Do not mix SI units: distances in meters, charge in coulombs.
K, u_E	Kinetic energy (J); field energy density (J/m^3)	Use $K_i+U_i=K_f+U_f$ for conservative motion. Use $u_E=(1/2)\epsilon E^2$ for energy per field volume.

Choose a method: one point source -> Coulomb formulas; many source charges -> superposition; highly symmetric continuous charge -> Gauss law for E , then integrate E for ΔV ; capacitor circuit -> reduce series/parallel groups; energy question -> choose U from the given variables and conserve $K+U$.